

FORWARD RATES FUTURES

JÖRGEN BLOMVALL

ABSTRACT.

1. OIS

From the document interest rate instruments, it follows that

$$(1) \quad P_{float} = \sum_{(i,j,k) \in \mathcal{F}} N \left(\frac{d(\bar{T}_i)}{d(\bar{T}_0)} \frac{d(\bar{T}_k)}{d(\bar{T}_j)} - \frac{d(\bar{T}_k)}{d(\bar{T}_0)} \right),$$

$$(2) \quad P_{fix} = \sum_{k=1}^K N y t_k^c \frac{d(\tilde{T}_k)}{d(T_0)}.$$

For the quoted swap rates, the total value of the OIS that is paying fixed rate and receiving floating rate should be equal to zero (assuming $T_0 = \bar{T}_0$)

$$(3) \quad P = P_{float} - P_{fix} = 0 \Leftrightarrow$$

$$(4) \quad \sum_{k=1}^K N y t_k^c \frac{d(\tilde{T}_k)}{d(T_0)} = \sum_{(i,j,k) \in \mathcal{F}} N \left(\frac{d(\bar{T}_i)}{d(\bar{T}_0)} \frac{d(\bar{T}_k)}{d(\bar{T}_j)} - \frac{d(\bar{T}_k)}{d(\bar{T}_0)} \right) \Leftrightarrow$$

$$(5) \quad \sum_{k=1}^K N y t_k^c d(\tilde{T}_k) = \sum_{(i,j,k) \in \mathcal{F}} N \left(d(\bar{T}_i) \frac{d(\bar{T}_k)}{d(\bar{T}_j)} - d(\bar{T}_k) \right) \Leftrightarrow$$

$$(6) \quad y = \frac{\sum_{(i,j,k) \in \mathcal{F}} \left(d(\bar{T}_i) \frac{d(\bar{T}_k)}{d(\bar{T}_j)} - d(\bar{T}_k) \right)}{\sum_{k=1}^K t_k^c d(\tilde{T}_k)}$$

The pricing of the floating can be simplified by a telescope sum if $\bar{T}_{i_1} = \bar{T}_0$, $\bar{T}_{j_k} = \bar{T}_{k_k}$, $\bar{T}_{j_l} = \bar{T}_{i_{l+1}}$ and when $\bar{T}_{j_n} = \tilde{T}_K$, which gives

$$(7) \quad y = \frac{\sum_{(i,j,k) \in \mathcal{F}} \left(d(\bar{T}_i) \frac{d(\bar{T}_k)}{d(\bar{T}_j)} - d(\bar{T}_k) \right)}{\sum_{k=1}^K t_k^c d(\tilde{T}_k)} = \frac{d(\bar{T}_0) - d(\tilde{T}_K)}{\sum_{k=1}^K t_k^c d(\tilde{T}_k)}.$$

For instrument i and suitable choices of $o_{t,i,k}$, the discount factors at time t can be computed from the systematic risk factor states $x_t^s \in \mathbb{R}^{n_t^s \times 1}$ by

$$(8) \quad d(\tilde{T}_k) = e^{o_{t,i,k}^T x_t^s},$$

giving

$$(9) \quad g_{t,i}(x_t^s) = \frac{e^{o_{t,i,0}^T x_t^s} - e^{o_{t,i,K}^T x_t^s}}{\sum_{k=1}^K t_k^c e^{o_{t,i,k}^T x_t^s}}$$

with

$$(10) \quad \nabla_{x_t^s} g_{t,i}(x_t^s) = \frac{e^{o_{t,i,0}^T x_t^s} o_{t,i,0} - e^{o_{t,i,K}^T x_t^s} o_{t,i,K}}{\sum_{k=1}^K t_k^c e^{o_{t,i,k}^T x_t^s}} - \frac{e^{o_{t,i,0}^T x_t^s} - e^{o_{t,i,K}^T x_t^s}}{\left(\sum_{k=1}^K t_k^c e^{o_{t,i,k}^T x_t^s} \right)^2} \sum_{k=1}^K t_k^c e^{o_{t,i,k}^T x_t^s} o_{t,i,k}.$$

The systematic risk factors $x_t^s = (x_t^p; x_t^c)$ are split into the principal components, $x_t^p \in \mathbb{R}^{n_t^p \times 1}$, from PCA analysis of the interest rate curves, and central bank interest rate changes $x_t^c \in \mathbb{R}^{n_t^c \times 1}$ (steps in the forward rate curve). The vectors $o_{t,i,k}$ are dependent on the parameters θ^g and the vector valued function for all the instruments can be described by $g_t(\theta^g, x_t^s)$, where the dependence on θ^g can be described by $o_{t,i,k} = \begin{pmatrix} O_{t,i,k} \theta^g \\ (2) \\ o_{t,i,k} \end{pmatrix}$ for

appropriate choices of $O_{t,i,k} \in \mathbb{R}^{n_t^p \times n^g}$.

2. TERM STRUCTURE MODELING

The state space, $x_t = (x_t^s; x_t^u) \in \mathbb{R}^{n_t^x \times 1}$ is split into the systematic risk factor states $x_t^s \in \mathbb{R}^{n_t^s \times 1}$, which define the interest rate curve, and the unsystematic risk factor states $x_t^u \in \mathbb{R}^{n_t^u \times 1}$, which define price deviations for each instrument. States evolve as

$$(11) \quad x_t = F_t(\theta^F) x_{t-1} + w_t,$$

where $F_t(\theta^F) \in \mathbb{R}^{n_t^x \times n_{t-1}^x}$ in many cases is a diagonal matrix with positive values smaller than (or equal to) one, which (can) drive x_t towards zero and $w_t \in \mathbb{R}^{n_t^x \times 1}$ is the noise. Note that for dates when a step is removed or introduced, then $F_t(\theta^F)$ will not be diagonal, which can be dealt with by multiplying $\text{diag}(\theta^F)$ with the predetermined matrices A_t and B_t ,

$$(12) \quad F_t(\theta^F) = A_t \text{diag}(\theta^F) B_t.$$

It is assumed that there is covariation in the term structure is described by some parameters $\theta^g \in \mathbb{R}^{n^g \times 1}$. Observations of the market prices $z_t \in \mathbb{R}^{n_t^z \times 1}$,

$$(13) \quad z_t = g_t(\theta^g, x_t^s) + I_t^z x_t^u + v_t,$$

has different number of observations, n_t^z , over time and the nonlinear function $g_t : \mathbb{R}^{(n^g + n_t^s) \times 1} \rightarrow \mathbb{R}^{n_t^z \times 1}$ determines the quoted rates (e.g. interest rates) for instruments, $I_t^z \in \mathbb{R}^{n_t^z \times n_t^u}$ contains ones for the instruments which have an observation at time t , and $v_t \in \mathbb{R}^{n_t^z \times 1}$ is noise.

The log-likelihood optimization problem is given by

$$(14) \quad \max \ln f_0^x(x_0) + \sum_{t=1}^T (\ln f_t^w(w_t) + \ln f_t^v(v_t)),$$

$$(15) \quad \text{s.t. } x_t = F_t(\theta^F) x_{t-1} + w_t, \quad t = 1, \dots, T,$$

$$(16) \quad z_t = g_t(\theta^g, x_t^s) + I_t^z x_t^u + v_t, \quad t = 1, \dots, T,$$

where $f_t^w : \mathbb{R}^{n_t^x \times 1} \rightarrow 1$ and $f_t^v : \mathbb{R}^{n_t^z \times 1} \rightarrow 1$ are likelihood functions. Note that if g_t is a linear function, f_t^w and f_t^v are multivariate normal distributions, then the optimal solution is given by the Rauch-Tung-Striebel version of the Kalman filter. Given these assumptions, the objective is quadratic and the constraints linear, using the Riccati recursion, the optimal solution can be found by a single forward and reverse pass. The standard Kalman filter should be found by sequentially solving

$$(17) \quad \max \ln f_t^w(w_t) + \ln f_t^v(v_t),$$

$$(18) \quad \text{s.t. } x_t = F_t(\theta^F) x_{t-1} + w_t,$$

$$(19) \quad z_t = g_t(\theta^g, x_t^s) + I_t^z x_t^u + v_t,$$

for each $t = 1, \dots, T$. When at least one assumption does not hold, then quadratic subproblems can be solved iteratively to find a local optima.

If the random variables and initial state have the distributions

$$(20) \quad v_t \sim N(0, G_t \Sigma^v G_t^\top), \quad t = 1, \dots, T,$$

$$(21) \quad w_t \sim N(0, D_t \Sigma^w D_t^\top), \quad t = 1, \dots, T,$$

$$(22) \quad x_0 \sim N(a^x, \Sigma^0),$$

then the all the parameters can be denoted $\theta = (\theta^F, \Sigma^w, \Sigma^v, a^x, \Sigma^0, \theta^g)$.

2.1. Parameter estimation.

- (23) $A_t = I,$
- (24) $B_t \approx I$ rows are removed when central bank effective date has passed,
- (25) $D_t = I,$
- (26) $I_t^z \approx I$ 0 if missing OIS observation,
- (27) $G_t = I_t^z$ 0 if missing OIS observation,
- (28) $\Sigma^v = \text{diagonal matrix},$
- (29) $\Sigma^w = \text{diagonal matrix},$
- (30) $F_t(\theta^F) = A_t \text{diag}(\theta^F) B_t$
- (31) $x_0^0 = a^x,$
- (32) $P_0^0 = \Sigma^0,$

- (33) $x_t = F_t(\theta^F) x_{t-1} + w_t, \quad t = 1, \dots, T,$
- (34) $z_t = g_t(\theta^g, x_t^s) + I_t^z x_t^u + v_t, \quad t = 1, \dots, T,$
- (35) $v_t \sim N(0, G_t \Sigma^v G_t^\top), \quad t = 1, \dots, T,$
- (36) $w_t \sim N(0, D_t \Sigma^w D_t^\top), \quad t = 1, \dots, T,$
- (37) $x_0 \sim N(a^x, \Sigma^0), \quad t = 1, \dots, T,$

The nonlinear constraint is linearized

$$(38) \quad g_t(\theta^g, x_t^s) \approx g_t(\theta^g, x_t^{t-1,s}) + \nabla_{x^s} g_t(\theta^g, x_t^{t-1,s})(x_t^s - x_t^{t-1,s})$$

Given the initial conditions

$$(39) \quad x_0^0 = a^x,$$

$$(40) \quad P_0^0 = \Sigma^0,$$

then the vectors and matrices in the predictor step can for $t = 1, \dots, T$ be computed by

$$(41) \quad x_t^{t-1} = F_t(\theta^F) x_{t-1}^{t-1} = A_t \text{diag}(\theta^F) B_t x_{t-1}^{t-1},$$

$$(42) \quad P_t^{t-1} = F_t(\theta^F) P_{t-1}^{t-1} F_t(\theta^F)^\top + D_t \Sigma^w D_t^\top$$

$$(43) \quad = A_t \text{diag}(\theta^F) B_t P_{t-1}^{t-1} B_t^\top \text{diag}(\theta^F) A_t^\top + D_t \Sigma^w D_t^\top$$

and the update step for $t = 1, \dots, T$ by

$$\begin{aligned}
(44) \quad H_t^{t-1}(\theta^g) &= [\nabla_{x^s} g_t(\theta^g, x_t^{t-1,s}) \ I_t^z], \\
(45) \quad u_t^{t-1}(\theta^g) &= g_t(\theta^g, x_t^{t-1,s}) - \nabla_{x^s} g_t(\theta^g, x_t^{t-1,s}) x_t^{t-1,s}, \\
(46) \quad \epsilon_t &= z_t - H_t^{t-1}(\theta^g) x_t^{t-1} - u_t^{t-1}(\theta^g) \\
(47) \quad &= z_t - \nabla_{x^s} g_t(\theta^g, x_t^{t-1,s}) x_t^{t-1,s} - I_t^z x_t^{t-1,u} - g_t(\theta^g, x_t^{t-1,s}) \\
(48) \quad &\quad + \nabla_{x^s} g_t(\theta^g, x_t^{t-1,s}) x_t^{t-1,s} \\
(49) \quad &= z_t - I_t^z x_t^{t-1,u} - g_t(\theta^g, x_t^{t-1,s}), \\
(50) \quad S_t &= H_t^{t-1}(\theta^g) P_t^{t-1} H_t^{t-1}(\theta^g)^\top + G_t \Sigma^v G_t^\top, \\
(51) \quad K_t &= P_t^{t-1} H_t^{t-1}(\theta^g)^\top S_t^{-1} \Leftrightarrow \\
(52) \quad K_t^T &= S_t^{-1} H_t^{t-1}(\theta^g) P_t^{t-1}, \\
(53) \quad x_t^t &= x_t^{t-1} + K_t \epsilon_t, \\
(54) \quad P_t^t &= (I - K_t H_t^{t-1}(\theta^g)) P_t^{t-1} \\
(55) \quad &= P_t^{t-1} - K_t H_t^{t-1}(\theta^g) P_t^{t-1} \\
(56) \quad &= P_t^{t-1} - P_t^{t-1} H_t^{t-1}(\theta^g)^\top S_t^{-1} H_t^{t-1}(\theta^g) P_t^{t-1}.
\end{aligned}$$

With $S_t = \Sigma_t^\epsilon$, the log-likelihood

$$(57) \quad l(\theta; \epsilon_{1:T}(\theta)) = -\frac{1}{2} \sum_{t=1}^T (n_t^z \ln 2\pi + \ln |\Sigma_t^\epsilon| + \epsilon_t^\top (\Sigma_t^\epsilon)^{-1} \epsilon_t),$$

where $\ln |\Sigma_t^\epsilon| = 2 \sum_i \ln L_{t,i,i}$ can be efficiently computed with the Cholesky factorization $L_t L_t^\top = \Sigma_t^\epsilon$.

3. NOISY OPTIMIZATION

With Automatic Differentiation, the objective and gradient can efficiently be computed. However, due to the scale of the computations, noise appear in the computed values due to the limited numerical precision of computers.

Suppose that the function $f(x) : \mathbb{R}^{n \times 1} \rightarrow \mathbb{R}^{1 \times 1}$ only have noisy objective function values, $f_i \approx f(x_i)$ and gradients $g_i \approx \nabla_x f(x_i)$ that have been evaluated in points $x_i \in \mathbb{R}^{n \times 1}$ for $i = 1, \dots, m$. Define the second order Taylor approximations,

$$(58) \quad f(\bar{x} + \Delta x) \approx \frac{1}{2} \Delta x^T Q \Delta x + q^T \Delta x + c,$$

of the objective function values. These are applied locally for $k \in \mathcal{K} = \{1, \dots, m\}$ using the points $i \in \mathcal{I}_k \subseteq \mathcal{K}$ to determine

$$(59) \quad f_i = \frac{1}{2} \Delta x_{k,i}^T Q_k \Delta x_{k,i} + q_k^T \Delta x_{k,i} + c_k + e_i + \xi_{k,i},$$

$$(60) \quad g_i = Q_k \Delta x_{k,i} + q_k + \bar{e}_i + \bar{\xi}_{k,i},$$

where $\Delta x_{k,i} = x_i - x_k$, e_i is the error in objective function value (due to noise), $\xi_{k,i}$ captures the error in objective function value introduced by the Taylor approximation, $\bar{e}_i$ is the error in gradient (due to noise) and $\bar{\xi}_{k,i}$ captures the error in gradient introduced

by the Taylor approximation. The objective is

$$(61) \quad \min_{Q_k, q_k, c_k, e_i, \xi_{k,i}, \bar{e}_i, \bar{\xi}_{k,i}} h = \frac{1}{2} \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{I}_k} \left(w_{k,i}^\xi \xi_{k,i}^2 + \bar{w}_{k,i}^\xi \bar{\xi}_{k,i}^T \bar{\xi}_{k,i} \right) + \frac{1}{2} \sum_{i=1}^m (w_i^e e_i^2 + \bar{w}_i^e \bar{e}_i^T \bar{e}_i),$$

where e.g.

$$(62) \quad w_{k,i}^\xi = \frac{1}{\sigma^2},$$

$$(63) \quad \bar{w}_{k,i}^\xi = \frac{1}{n \bar{\sigma}^2},$$

$$(64) \quad w_i^e = 1,$$

$$(65) \quad \bar{w}_i^e = 1,$$

can be used.

The total number of variables from each quadratic approximation is $n^2 + n + 1$, which gives the total number of variables

$$(66) \quad \underbrace{m(n^2 + n + 1)}_{\frac{1}{2} \Delta x_{k,i}^T Q_k \Delta x_{k,i} + q_k^T \Delta x_{k,i} + c_k} + m(\underbrace{1}_{e_i} + \underbrace{n}_{\bar{e}_i}) + \sum_{k=1}^m |\mathcal{I}_k| (\underbrace{1}_{\xi_{k,i}} + \underbrace{n}_{\bar{\xi}_{k,i}}).$$

Since $|\mathcal{I}_k| \leq m$, an upper bound on the number of variables can be determined by

$$(67) \quad m(n^2 + n + 1) + m(1 + n) + mn(1 + n) = m(2n^2 + 3n + 2),$$

which is more than the number of observations $m(1 + n)$ of (f_i, g_i) .

3.1. Draft for determining optimal solutions. Let $\bar{y}_k = (\text{vec}(Q_k); q_k; c_k)$ with $\bar{a}_{i,k} = (\text{vec}(\frac{1}{2} \Delta x_{k,i} \Delta x_{k,i}); \Delta x_{k,i}; 1)$ and $\bar{A}_{i,k} = (\tilde{A}_{i,k}; I; 0)$, where

$$(68) \quad \tilde{A}_{i,k} = \begin{pmatrix} \Delta x_{k,i}^T & 0 & 0 & 0 \\ 0 & \Delta x_{k,i}^T & 0 & 0 \\ \vdots & 0 & \ddots & 0 \\ 0 & 0 & 0 & \Delta x_{k,i}^T \end{pmatrix},$$

then the constraints can be expressed as

$$(69) \quad f_i = \bar{a}_{i,k}^T \bar{y}_k + e_i + \xi_{k,i},$$

$$(70) \quad g_i = \bar{A}_{i,k} \bar{y}_k + \bar{e}_i + \bar{\xi}_{k,i},$$

since Q_k is a symmetric matrix the unique elements can be expressed with $\text{vech}(Q_k)$, and all elements can be expressed with $\text{vec}(Q_k) = D_k \text{vech}(Q_k)$, where D_k is the **duplication matrix**. With $y_k = (\text{vech}(Q_k); q_k; c_k)$ with $a_{i,k} = (D_k^T \text{vec}(\frac{1}{2} \Delta x_{k,i} \Delta x_{k,i}); \Delta x_{k,i}; 1)$ and $A_{i,k} = (\tilde{A}_{i,k} D_k; I; 0)$ the constraints can be reformulated to

$$(71) \quad f_i = a_{i,k}^T y_k + e_i + \xi_{k,i},$$

$$(72) \quad g_i = A_{i,k} y_k + \bar{e}_i + \bar{\xi}_{k,i}.$$

Note that

$$(73) \quad h_{1,i,k} = \frac{1}{2}(f_i - a_{i,k}^T y_k - e_i)^2 = \frac{1}{2}f_i^2 - f_i a_{i,k}^T y_k - f_i e_i + \frac{1}{2}y_k^T a_{i,k} a_{i,k}^T y_k + e_i a_{i,k}^T y_k + e_i^2$$

$$(74) \quad h_{2,i,k} = \frac{1}{2}(g_i - A_{i,k} y_k - \bar{e}_i)^T (g_i - A_{i,k} y_k - \bar{e}_i)$$

$$(75) \quad = \frac{1}{2}g_i^T g_i - g_i^T A_{i,k} y_k - g_i^T \bar{e}_i + \frac{1}{2}y_k^T A_{i,k}^T A_{i,k} y_k + \bar{e}_i^T A_{i,k} y_k + \frac{1}{2}\bar{e}_i^T \bar{e}_i$$

and eliminating $\xi_{k,i}$ and $\bar{\xi}_{k,i}$ from the objective gives

$$(76) \quad h = \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{I}_k} \left(w_{k,i}^\xi h_{1,i,k} + \bar{w}_{k,i}^\xi h_{2,i,k} \right) + \frac{1}{2} \sum_{i=1}^m (w_i^e e_i^2 + \bar{w}_i^e \bar{e}_i^T \bar{e}_i)$$

$$(77) \quad = \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{I}_k} w_{k,i}^\xi \left(\frac{1}{2}f_i^2 - f_i a_{i,k}^T y_k - f_i e_i + \frac{1}{2}y_k^T a_{i,k} a_{i,k}^T y_k + e_i a_{i,k}^T y_k + e_i^2 \right)$$

$$(78) \quad \sum_{k \in \mathcal{K}} \sum_{i \in \mathcal{I}_k} \bar{w}_{k,i}^\xi \left(\frac{1}{2}g_i^T g_i - g_i^T A_{i,k} y_k - g_i^T \bar{e}_i + \frac{1}{2}y_k^T A_{i,k}^T A_{i,k} y_k + \bar{e}_i^T A_{i,k} y_k + \frac{1}{2}\bar{e}_i^T \bar{e}_i \right)$$

$$(79) \quad + \frac{1}{2} \sum_{i=1}^m (w_i^e e_i^2 + \bar{w}_i^e \bar{e}_i^T \bar{e}_i)$$

$$(80) \quad = \sum_{k \in \mathcal{K}} \left(\bar{c}_k - \alpha_k^T y_k + \frac{1}{2}y_k^T \bar{S}_k y_k \right) + \frac{1}{2} \sum_{i=1}^m (w_i^e e_i^2 + \bar{w}_i^e \bar{e}_i^T \bar{e}_i)$$

where

$$(81) \quad \bar{c}_k = \sum_{i \in \mathcal{I}_k} \frac{1}{2} \left(w_{k,i}^\xi f_i^2 + \bar{w}_{k,i}^\xi g_i^T g_i \right) + \sum_{i \in \mathcal{I}_k} w_{k,i}^\xi f_i e_i + \sum_{i \in \mathcal{I}_k} \bar{w}_{k,i}^\xi g_i^T \bar{e}_i$$

$$(82) \quad \alpha_k = \sum_{i \in \mathcal{I}_k} \left(w_{k,i}^\xi (f_i - e_i) a_{i,k} + \bar{w}_{k,i}^\xi A_{i,k}^T (g_i - \bar{e}_i) \right)$$

$$(83) \quad \bar{S}_k = \sum_{i \in \mathcal{I}_k} \left(w_{k,i}^\xi a_{i,k} a_{i,k}^T + \bar{w}_{k,i}^\xi A_{i,k}^T A_{i,k} \right)$$

$$(84) \quad w_i = w_i^e + \sum_{k \in \mathcal{K}} w_{k,i}^\xi$$

$$(85) \quad \bar{w}_i = \bar{w}_i^e + \sum_{k \in \mathcal{K}} \bar{w}_{k,i}^\xi$$

The optimal y_k can be determined as a function of e_i and $\bar{e}_i$ by the stationary point

$$(86) \quad \nabla_{y_k} h = -\alpha_k + \bar{S}_k y_k = 0 \Leftrightarrow y_k = \bar{S}_k^{-1} \alpha_k,$$

which gives the objective

$$(87) \quad h = \sum_{k \in \mathcal{K}} \left(\bar{c}_k - \alpha_k^T y_k + \frac{1}{2} y_k^T \bar{S}_k y_k \right) + \frac{1}{2} \sum_{i=1}^m (w_i e_i^2 + \bar{w}_i \bar{e}_i^T \bar{e}_i)$$

$$(88) \quad = \sum_{k \in \mathcal{K}} \left(\bar{c}_k - \alpha_k^T \bar{S}_k^{-1} \alpha_k + \frac{1}{2} \alpha_k^T \bar{S}_k^{-T} \bar{S}_k \bar{S}_k^{-1} \alpha_k \right) + \frac{1}{2} \sum_{i=1}^m (w_i e_i^2 + \bar{w}_i \bar{e}_i^T \bar{e}_i)$$

$$(89) \quad = \sum_{k \in \mathcal{K}} \left(\bar{c}_k - \frac{1}{2} \alpha_k^T \bar{S}_k^{-1} \alpha_k \right) + \frac{1}{2} \sum_{i=1}^m (w_i e_i^2 + \bar{w}_i \bar{e}_i^T \bar{e}_i),$$

since $\bar{S}_k$ is symmetric.

The optimum with respect to the remaining variables e_i and $\bar{e}_i$ can be determined by the stationary point

$$(90) \quad \frac{\partial h}{\partial e_i} = \sum_{k \in \mathcal{K}} \left(\frac{\partial \bar{c}_k}{\partial e_i} - \alpha_k^T \bar{S}_k^{-1} \frac{\partial \alpha_k}{\partial e_i} \right) + w_i e_i = 0,$$

$$(91) \quad \nabla_{\bar{e}_i} h = \sum_{k \in \mathcal{K}} (\nabla_{\bar{e}_i} \bar{c}_k - \nabla_{\bar{e}_i}^T \alpha_k \bar{S}_k^{-1} \alpha_k) + \bar{w}_i \bar{e}_i = 0,$$

where

$$(92) \quad \frac{\partial \bar{c}_k}{\partial e_i} = w_{k,i}^\xi f_i,$$

$$(93) \quad \frac{\partial \alpha_k}{\partial e_i} = -w_{k,i}^\xi a_{i,k},$$

$$(94) \quad \nabla_{\bar{e}_i} \bar{c}_k = \bar{w}_{k,i}^\xi g_i,$$

$$(95) \quad \nabla_{\bar{e}_i} \alpha_k = -\bar{w}_{k,i}^\xi A_{i,k}^T.$$

This gives the equation system

$$(96) \quad \frac{\partial h}{\partial e_i} = \sum_{k \in \mathcal{K}} \left(w_{k,i}^\xi f_i + w_{k,i}^\xi a_{i,k}^T \bar{S}_k^{-1} \alpha_k \right) + w_i e_i = 0,$$

$$(97) \quad \nabla_{\bar{e}_i} h = \sum_{k \in \mathcal{K}} \left(\bar{w}_{k,i}^\xi g_i + \bar{w}_{k,i}^\xi A_{i,k} \bar{S}_k^{-1} \alpha_k \right) + \bar{w}_i \bar{e}_i = 0,$$

where α_k contain e_k and $\bar{e}_k$. Hence, there is no efficient analytical solution to determine e_i and $\bar{e}_i$ other than solving the full equation system. However, one possibility is to use a fixed point iteration to obtain the values.

REFERENCES

(J. Blomvall) DEPARTMENT OF MANAGEMENT AND INDUSTRIAL ENGINEERING, LINKÖPINGS UNIVERSITET, SE-58183 LINKÖPING, SWEDEN. PHONE: +46 13 281406, FAX: +46 13 285770

Email address: jorgen.blomvall@liu.se